Lethe CFD-DEM: problems and comparison with our solver

EN 649 · Kaustubh Ashok (24B1522) · gas-fluidized bed, minimum fluidization velocity u_mf · Liu & van Wachem (2019) benchmark (paper simulation 1.00 m/s, NETL experiment 1.05 m/s)

1. Summary

We moved from our own Python/Numba DEM solver (gas treated through a 1-D volume-flux model, "quasi-two-way") to Lethe v1.2.0, an open-source finite-element CFD-DEM code, so that the gas momentum equation is actually solved with Gidaspow drag. Getting there took far more effort than expected. Of the 14 problems below, 8 were our own setup or convention mismatches, 3 are properties of Lethe, and 3 are mixed. Two of them invalidated results already obtained (a zero-velocity inlet, and a factor 2.25 error in gas pressure drop).

Where things stand after fixing what could be fixed (5 beds, patched Lethe):

- Fluidization onset on the way up is reproduced: **0.96 m/s**, against 1.00 m/s (paper) and 1.05 m/s (experiment).
- u_mf from the descending branch is **1.14 m/s** (1.28 m/s if forced to the bed weight), i.e. 14% above our quasi-two-way model and 8% above the experiment.
- The plateau pressure drop is **723 Pa, 80% of the 905 Pa bed weight**, in every bed. Our solver holds it at 100%. So Lethe fails our Checkpoint 3 (±10% of bed weight).
- It needed a **patch to Lethe's Gidaspow drag** to get the right fixed-bed pressure drop. The patch is ours and is not confirmed with the Lethe developers.

2. The two setups side by side

	Our solver (quasi-two-way)	Lethe v1.2.0 (patched)
Particles	2-D monolayer discs, symplectic Euler, dt = 3.328 μs	3-D spheres in a pseudo-2-D slab (depth 1.0005 d_p), velocity-Verlet, dt = 3.3 μs
Contact	Linear spring-dashpot-slider, kt/kn = 2/7, no rotation of spheres	Linear contact, kt/kn fixed at 0.4, spheres can spin; Young's modulus tuned so kn = 3654 N/m

	Our solver (quasi-two-way)	Lethe v1.2.0 (patched)
Gas model	1-D volume flux per row: u_g = (U_g − q_s)/ε, **gas momentum not solved**	Finite-element volume-averaged Navier-Stokes (VANS model A), gas momentum solved in 3-D
Drag	Gidaspow (Ergun ε < 0.8, Wen-Yu above), total-force form	Gidaspow, Lethe's total-force coefficient, patched for model A
Void fraction	Slab voidage on 24 × 56 cells (3 d_p), refreshed every N DEM steps	PCM L2 projection on the same 24 × 56 mesh (+3 plenum cells)
Inlet	Superficial velocity U_g directly	Interstitial velocity only, so a plenum and a floating floor wall are needed
Ramp	0 to peak and back (5:1 update ratio used for the main results)	0 → 1.6 → 0 m/s at 1 m/s per second, fluid step 1.98 ms (1.32 ms for seed 12)
Cost	Several hours per bed, can run several at once	About 1.1 s per fluid step; 50–70 min per bed on 2 cores, one at a time
Beds	14 beds (seed 23 excluded)	5 beds (seeds 7, 11–14)

Run times are not directly comparable: the ramp lengths and the update ratios differ.

3. Problems, in the order they appeared

Category: **Ours** = our setup error; **Convention** = difference between how Lethe and our model define something; **Lethe** = behaviour of the code itself.

#	Problem	Symptom	Cause	Fix / status	Category
1	Installing Lethe on Windows	No native Windows build	Lethe targets Linux	WSL2 + Ubuntu 24.04 + Docker, 12 GB container image. The admin step had to be run by the user.	Lethe
2	Input files rejected	Parser errors on first runs	Python on Windows wrote CRLF line endings	Write with LF (`newline="\n"`)	Ours
3	Run refuses to start	"no boundary condition for id"	Lethe needs a BC for every boundary id	Slip BC on the four side faces (number = 6)	Lethe

#	Problem	Symptom	Cause	Fix / status	Category
4	No true 2-D	2-D void fraction is areal (area, not volume), so it is unusable for a 2-D bed	Lethe's 2-D mode is cross-section based	Run in 3-D as a pseudo-2-D slab of depth ≈ d_p so ε = 1 − ΣV_p/(Δx Δy d_p), as in our model	Lethe
5	Inlet velocity applied as interstitial	Effective superficial velocity was ε·U_in, so comparisons at "U_g = 0.5" were invalid	Velocity BC is the gas velocity, not the flux	Added a 3-cell empty plenum and a floating wall for the floor. The earlier PCM comparison was thrown away.	Convention
6	Contact model not identical	Cannot match kt/kn or the 2-D rotation constraint	Lethe fixes kt/kn = 0.4 and spheres rotate	Young's modulus solved so kn(Lethe) = 3654 N/m (pp 3.0e7 Pa, pw 1.8e7 Pa). Residual difference accepted.	Convention
7	Time-dependent inlet silently ignored	First pilot ramp (72 min) gave ΔP ≈ 0 Pa the whole time	A function BC stays at its t = 0 value unless `set time dependent = true`	Flag added. The first pilot was invalid.	Lethe
8	**Gas pressure drop 2.25× too high**	482 Pa at 0.5 m/s in the settled bed; Ergun gives ≈ 214 Pa	Lethe's Gidaspow per-particle coefficient is in total-force form (1/ε² dense branch, ε^−3.65 dilute), but model A balances drag against ε∇p, so ΔP = Ergun/ε (214/0.446 ≈ 480)	Switching to model B (517 Pa), void fraction qcm (457 Pa) or spm (475 Pa), or turning pressure and buoyancy forces off (483 Pa) did not help. Fixed with a one-line patch (coefficient × local ε) compiled from vans_assemblers.cc and loaded by LD_PRELOAD: 204.5 Pa. Not verified against the Lethe developers.	Lethe / Convention
9	Early fluidization	With the unpatched code, ΔP reached the bed weight at ≈ 0.65 m/s (ours ≈ 1.0)	Consequence of #8	Removed by the patch	Lethe

#	Problem	Symptom	Cause	Fix / status	Category
10	Solver divergence	Fluid solve blew up (ΔP → 10^130 Pa) at t = 2.24 s in the pilot and at 1.98 s for seed 12, both on the descending branch	Nonlinear fluid solve failing in one step; cause not identified	Peak lowered to 1.6 m/s; seed 12 rerun with a 1.32 ms fluid step. Sporadic, so it can recur.	Lethe
11	Plateau at 80% of bed weight	723 ± 2 Pa against 905 Pa in all five beds	The bed lifts, splits (ε ≈ 0.84 at mid-height) and drops onto the floor; instantaneous ΔP swings 464–754 Pa within 0.6 s. Where the bed lands the floor carries part of the weight, so the time mean is low. Wall friction was ruled out (671 Pa with friction, 661 Pa without).	Not resolved. Our diagnosis rests on three field snapshots.	Lethe (not diagnosed)
12	Our u_mf extractor cannot be applied	It needs the plateau above 90% of the bed weight	#11	u_mf reported two ways: measured plateau (1.14 m/s) and plateau forced to 905 Pa (1.28 m/s)	Ours
13	Our own tooling mistakes	Output of one variant overwrote the pilot's output directory; a spec parser split on commas inside inlet expressions; Windows-to-WSL quoting; a UTF-8 character broke an analysis script	Scripts written under time pressure	Fixed in `variants.py` and the driver scripts	Ours
14	Cost and scheduling	≈ 1.1 s per fluid step, 2 cores, one bed at a time; a PC restart and a lost session each killed a batch earlier	Single-machine budget	Detached drivers with per-run logs	Mixed

4. Lethe on its own

Things that hold regardless of what we compare against.

- **Gidaspow drag in v1.2.0 is internally inconsistent with model A.** The per-particle coefficient is the total-force form, but the gas momentum equation uses ε∇p. Without the patch, any fluidized-bed pressure drop it reports is too high by about 1/ε. This is our reading of the source plus a numerical match (480 Pa predicted, 482 Pa measured); we have not checked it with the developers.
- **Time-dependent boundary conditions are easy to get wrong silently.** There is no warning when the flag is missing; the run just proceeds at the t = 0 value.
- **No usable 2-D void fraction for a monolayer.** The 2-D mode is areal, so slab-like cases need a 3-D mesh.
- **Inlet is a gas velocity.** A superficial-velocity inlet needs a plenum.
- **Cost.** About 1.1 s per fluid step on 2 cores means 50–70 minutes for a short 3.2 s ramp.
- **Robustness.** Two divergences in six ramps on the descending branch, with no error message beyond `std::exception`.
- **Tooling.** The run needs Linux (WSL2 + Docker) and a 12 GB image; patching a single file took about 30 seconds of compile time once the source was fully extracted.

5. Results

All beds: seed-based settled beds from our DEM, re-settled in Lethe, ramp 0 → 1.6 → 0 m/s. u_mf from the descending branch: quadratic fit to ΔP(U_g) for 0.2 < U_g < 0.9, intersected with the plateau.

Seed	Plateau (up branch, 1.15–1.5 m/s)	u_mf, measured plateau (m/s)	u_mf, plateau forced to 905 Pa (m/s)	Onset on the way up (m/s)
7	727 Pa (80%)	1.147	1.292	0.97
11	723 Pa (80%)	1.130	1.275	0.96
12 (fluid step 1.32 ms)	720 Pa (80%)	1.142	1.292	0.94
13	721 Pa (80%)	1.147	1.298	0.98
14	722 Pa (80%)	1.122	1.267	0.94
Mean ± SD	**723 ± 3 Pa**	**1.138 ± 0.010**	**1.285 ± 0.013**	**0.96 ± 0.02**

Comparison:

Source	u_mf (m/s)	Plateau / bed weight
Paper simulation (Liu & van Wachem)	1.00	about 100%
NETL experiment	1.05	about 100%
Our solver, one-way 5:1	see README	100% (104 of 105 runs pass)
Our solver, quasi-two-way 5:1 (14 beds)	0.990 ± 0.009	100%
Lethe, patched, onset on the way up	0.96 ± 0.02	—
Lethe, patched, descending branch, measured plateau	1.14 ± 0.01	80%
Lethe, patched, descending branch, plateau forced to 905 Pa	1.28 ± 0.01	forced
Lethe, unpatched (seed 7 pilot)	onset ≈ 0.65	77–86%

6. What this does and does not show

- The gas drag and pressure drop are now consistent with Ergun at fixed bed (204.5 Pa against 214 Pa at 0.5 m/s), and the fluidization onset is in the right place. That supports the drag calibration in our own model.
- The Lethe u_mf is not a clean number. It depends on how the plateau is defined, and the plateau itself is 20% below the bed weight for a reason we have only partly explained.
- Bed-to-bed spread is small (0.01 m/s), comparable to our model, so the result is repeatable even though it is offset.
- Only five beds, all with the same ramp and a single mesh. The seed 12 fluid step differs.
- The patch is not an independent confirmation: it was derived from the same mismatch that it removes, and checked only on a fixed bed.

7. Options from here

1. **Report as is.** Lethe as a cross-check: right onset, offset descending-branch u_mf, 80% plateau, with the limitations above.

2. **Diagnose the 80% plateau.** Run longer holds at fixed U_g, count particles resting on the floor, and test a finer mesh or a smaller fluid step on one bed. Likely a day of work.
3. **Stay with the quasi-two-way model as the main result** and put Lethe in the README as a documented cross-check with the patch and scripts under a `lethe/` folder.

Recommendation: option 3, with option 2 only if the 80% plateau is considered important to resolve before submission.